

Moledi Events — UML Structure

Class Diagrams (DC-01 → DC-10) + Component Diagram

Conventions

- All field names in **English**
- `user_id` → references `User` table (no separate Organizer table)
- `?` = nullable field
- `[]` = array / list of values
- Commission lookup order: `UserCommissionConfig` (per user) → `CommissionConfig` (global)

DC-01 — Identity & Session MVP

Visitor

Field	Type
visitor_id	UUID PK
browser	String
os	String
language	String
screen_resolution	String
ip_hashed	String
account_id	UUID? FK→User
created_at	DateTime

PublicSession

Field	Type
session_id	UUID PK
visitor_id	UUID FK
ip_hashed	String
user_agent	String
created_at	DateTime

PublicAction

Field	Type
action_id	UUID PK
visitor_id	UUID FK
type	ActionType
entity_id	UUID
entity_type	String
status	ActionStatus
idempotency_key	String unique
short_id	String(8)
created_at	DateTime

GDPRConsent

Field	Type
consent_id	UUID PK
visitor_id	UUID FK
accepted	Boolean

timestamp	DateTime
policy_version	String
ip_hashed	String

RateLimit

Field	Type
ip_hashed	String PK
visitor_id	UUID PK
action_type	ActionType PK
counter	Integer
window_start	DateTime
blocked_until	DateTime?

Enums

```

ActionType = VOTE | TICKET_PURCHASE | DONATION | CF_CONTRIBUTION | CONTEST_ENTR
ActionStatus = PENDING | CONFIRMED | FAILED | EXPIRED | REFUNDED

```

Relations

- Visitor 1 → * PublicSession
- Visitor 1 → * PublicAction
- Visitor 1 → 0..1 GDPRConsent
- Visitor 1 → * RateLimit
- Visitor * → 0..1 User

DC-02 — Authentication & Users MVP

User

Field	Type
user_id	UUID PK
role	UserRole
full_name	String
email	String unique
email_verified	Boolean
phone	String?
phone_country_code	String?
password_hash	String?
avatar_url	String?
status	UserStatus
acquisition_source_id	UUID? FK
linked_visitor_id	UUID? FK
payout_phone	String?
payout_operator	String?
payout_phone_history	String[]
pseudonymised	Boolean
created_at	DateTime
deleted_at	DateTime?

UserPreferences

Field	Type
user_id	UUID PK FK
preferred_event_types	String[]
organization_frequency	String

activity_country	String
language	Language
notif_email_validation	Boolean
notif_email_rejection	Boolean
notif_email_payment	Boolean
notif_email_payout	Boolean
notif_email_ticket	Boolean
notif_whatsapp_validation	Boolean
notif_whatsapp_payment	Boolean
notif_whatsapp_payout	Boolean
newsletter_frequency	NewsletterFreq

UserSession

Field	Type
session_id	UUID PK
user_id	UUID FK
access_token	String
refresh_token	String
ip	String
browser	String
created_at	DateTime
expires_at	DateTime
invalidated	Boolean

Token

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Field	Type
token_id	UUID PK
user_id	UUID FK
value	String unique
type	TokenType
expires_at	DateTime
used	Boolean
created_at	DateTime

TwoFA

Field	Type
user_id	UUID PK FK
type	TwoFAType
totp_secret	String? encrypted
active	Boolean
activated_at	DateTime

LoginLog

Field	Type
log_id	UUID PK
user_id	UUID FK
ip	String
browser	String
success	Boolean
failed_attempts	Integer

blocked_until	DateTime?
created_at	DateTime

AdminPermission

Field	Type
user_id	UUID PK FK
view_events	Boolean
edit_events	Boolean
view_users	Boolean
manage_users	Boolean
view_finances	Boolean
manage_finances	Boolean
view_tickets	Boolean
reply_tickets	Boolean
mass_communication	Boolean
manage_pricing	Boolean
manage_aggregators	Boolean
feature_flags	Boolean
global_payout_block	Boolean
configure_payment_countries	Boolean

AcquisitionSource

Field	Type
source_id	UUID PK
name	String

type	SourceType
url	String?
active	Boolean
created_at	DateTime

Influencer

Field	Type
influencer_id	UUID PK
source_id	UUID FK
name	String
photo_url	String?
platform	String
followers_count	Integer?
notes	Text?
active	Boolean
users_referred	Integer
revenue_generated	Decimal

Enums

```

UserRole      = ORGANIZER | ADMIN | SUPER_ADMIN
UserStatus    = ACTIVE | SUSPENDED | DELETED
TokenType     = EMAIL_VERIFICATION | PASSWORD_RESET | JURY_INVITATION | ADMIN_IN
TwoFAType     = TOTP | EMAIL_OTP
Language      = FR | EN
NewsletterFreq = INSTANT | DAILY | WEEKLY | NEVER
SourceType    = SOCIAL_NETWORK | INFLUENCER | EVENT | ADVERTISING | OTHER

```

Relations

- User 1 → 1 UserPreferences

- User 1 → * UserSession
- User 1 → * Token
- User 1 → 0..1 TwoFA
- User 1 → * LoginLog
- User {ADMIN|SUPER_ADMIN} 1 → 1 AdminPermission
- User * → 0..1 AcquisitionSource
- AcquisitionSource 1 → * Influencer

DC-03 — Polls & Votes MVP

Poll

Field	Type
poll_id	UUID PK
user_id	UUID FK
slug	String unique
title	String
description	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
category	String
display_organizer_name	String
social_links	JSON
status	PollStatus
vote_type	VoteType

price_per_vote	Decimal?
vote_packs	JSON?
max_votes_per_visitor	Integer?
otp_enabled	Boolean
results_visibility	ResultsVisibility
show_grid_directly	Boolean
open_at	DateTime
close_at	DateTime
timezone	String
rejection_reason	String?
published_version	Integer
custom_fields	JSON?
created_at	DateTime

Candidate

Field	Type
candidate_id	UUID PK
poll_id	UUID FK
category_id	UUID? FK
real_name	String
display_name	String
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
biography	Text

phone	String? private
email	String? private
payout_phone	String?
score	Integer
rank	Integer
active	Boolean
position	Integer
custom_fields_data	JSON?

CandidateCategory

Field	Type
category_id	UUID PK
poll_id	UUID FK
name	String
position	Integer

Vote

Field	Type
vote_id	UUID PK
poll_id	UUID FK
candidate_id	UUID FK
visitor_id	UUID FK
user_id	UUID? FK
vote_type	VoteType

status	VoteStatus
verification_method	String
short_id	String(8)
ip_hashed	String
transaction_id	UUID? FK
created_at	DateTime

AntiDuplicate

Field	Type
ad_id	UUID PK
poll_id	UUID FK
method	AntiDupMethod
value_hashed	String
vote_id	UUID? FK
blocked_until	DateTime?
created_at	DateTime

VoteOTP

Field	Type
otp_id	UUID PK
poll_id	UUID FK
visitor_id	UUID FK
code_hashed	String
channel	OTPChannel
expires_at	DateTime

attempts	Integer
verified	Boolean
created_at	DateTime

UniqueCode

Field	Type
code_id	UUID PK
poll_id	UUID FK
value	String(8)
status	CodeStatus
vote_id	UUID? FK
used_at	DateTime?

JurySession

Field	Type
jury_id	UUID PK
poll_id	UUID FK
juror_email	String
invitation_token_id	UUID FK
jury_weight_percent	Integer
public_weight_percent	Integer
score_range_min	Integer
score_range_max	Integer
submitted	Boolean
created_at	DateTime

JuryScore

Field	Type
score_id	UUID PK
jury_id	UUID FK
candidate_id	UUID FK
score	Decimal
submitted_at	DateTime

WeightedScore

Field	Type
ws_id	UUID PK
poll_id	UUID FK
candidate_id	UUID FK
public_score	Integer
normalized_jury	Decimal
final_score	Decimal
calculated_at	DateTime

ClosingReport

Field	Type
report_id	UUID PK
poll_id	UUID FK
version	Integer
content_json	JSON
sha256_hash	String

pdf_url	String
public	Boolean
generated_at	DateTime
supervisor_id	UUID FK

PollNews

Field	Type
news_id	UUID PK
poll_id	UUID FK
title	String
body	Text
photos_urls	String[]
videos_urls	String[]?
published_at	DateTime

PollGallery

Field	Type
item_id	UUID PK
poll_id	UUID FK
media_url	String
media_type	MediaType
tag	GalleryTag
uploaded_at	DateTime

PollPartner

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Field	Type
partner_id	UUID PK
poll_id	UUID FK
name	String
logo_url	String?
other_images_urls	String[]?
website_url	String?
level	String?
validated	Boolean

PollFAQ

Field	Type
faq_id	UUID PK
poll_id	UUID FK
question	String
answer	Text
position	Integer

PollNotice

Field	Type
notice_id	UUID PK
poll_id	UUID FK
message	Text
type	NoticeType
permanent	Boolean

expires_at	DateTime?
published_at	DateTime

PollReport

Field	Type
report_id	UUID PK
poll_id	UUID FK
candidate_id	UUID? FK
reporter_name	String
reporter_phone	String
complaint_type	PollComplaintType
description	Text
attachments_urls	String[]
ticket_id	UUID? FK
created_at	DateTime

RefundPolicy

Field	Type
policy_id	UUID PK
campaign_id	UUID
campaign_type	String
refundable	Boolean
delay_hours	Integer?
percentage	Integer?

Enums

VoteType = FREE_VISITOR_ID | FREE_EMAIL | FREE_PHONE | FREE_SMS | FREE_W
PollStatus = DRAFT | PENDING_VALIDATION | PUBLISHED | OPEN | SUSPENDED | C
VoteStatus = COUNTED | NOT_COUNTED | CANCELLED | REFUNDED
AntiDupMethod = VISITOR_ID | EMAIL | PHONE | SMS_OTP | WHATSAPP_OTP | UNIQUE_
CodeStatus = UNUSED | USED | CANCELLED
OTPChannel = SMS | WHATSAPP
ResultsVisibility = PUBLIC | HIDDEN | AFTER_CLOSE
MediaType = PHOTO | VIDEO
GalleryTag = BEFORE | DURING | AFTER
NoticeType = ANNOUNCEMENT | POPUP
PollComplaintType = VOTE_NOT_COUNTED | PAYMENT_WITHOUT_VOTE | DUPLICATE_VOTE | TE

Relations

- Poll 1 → * Candidate
- Poll 1 → * CandidateCategory
- Candidate * → 0..1 CandidateCategory
- Poll 1 → * Vote
- Vote * → 1 Candidate
- Poll 1 → * AntiDuplicate
- Poll 1 → * VoteOTP
- Poll 1 → * UniqueCode
- Poll 1 → 0..1 JurySession
- JurySession 1 → * JuryScore
- Poll 1 → * WeightedScore
- Poll 1 → * ClosingReport
- Poll 1 → * PollNews
- Poll 1 → * PollGallery
- Poll 1 → * PollPartner
- Poll 1 → * PollFAQ
- Poll 1 → * PollNotice
- Poll 1 → * PollReport

- Poll 1 → 1 RefundPolicy

DC-04 — Events & Ticketing v1

Event

Field	Type
event_id	UUID PK
user_id	UUID FK
slug	String unique
title	String
description	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
category	String
status	EventStatus
mode	EventMode
start_at	DateTime
end_at	DateTime
timezone	String
visible_tabs	String[]
custom_fields	JSON?
created_at	DateTime

EventVenue

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Field	Type
venue_id	UUID PK
event_id	UUID FK
hall_name	String
address	String
city	String
country	String
google_maps_url	String?
floor_plan_url	String?
access_info	Text?

TicketType

Field	Type
ticket_type_id	UUID PK
event_id	UUID FK
name	String
category	TicketCategory
price	Decimal
total_stock	Integer
sold_count	Integer
reserved_count	Integer
perks	Text?
early_bird_ends_at	DateTime?
active	Boolean

PromoCode

Field	Type
promo_id	UUID PK
event_id	UUID FK
code	String
discount_percent	Integer
quota	Integer
used_count	Integer
valid_until	DateTime?
active	Boolean

TicketPurchase

Field	Type
purchase_id	UUID PK
event_id	UUID FK
ticket_type_id	UUID FK
visitor_id	UUID FK
user_id	UUID? FK
transaction_id	UUID FK
status	TicketStatus
qr_code_content	String unique
pdf_url	String?
delivery_channel	DeliveryChannel
buyer_name	String

buyer_email	String
buyer_phone	String?
custom_fields_answers	JSON?
promo_code_id	UUID? FK
short_id	String(8)
is_free_registration	Boolean
created_at	DateTime

TemporaryReservation

Field	Type
reservation_id	UUID PK
ticket_type_id	UUID FK
visitor_id	UUID FK
quantity	Integer
expires_at	DateTime
created_at	DateTime

EntryScan

Field	Type
scan_id	UUID PK
purchase_id	UUID FK
event_id	UUID FK
scanned_at	DateTime
operator_id	UUID FK
method	ScanMethod

scan_result	ScanResult
offline	Boolean

ProgramSession

Field	Type
session_id	UUID PK
event_id	UUID FK
title	String
description	Text?
start_at	DateTime
end_at	DateTime
room	String?
speaker_id	UUID? FK
day_number	Integer

Speaker

Field	Type
speaker_id	UUID PK
event_id	UUID FK
name	String
title	String
bio	Text?
photo_url	String?
additional_photos_urls	String[]?
social_links	JSON?

EventSponsor

Field	Type
sponsor_id	UUID PK
event_id	UUID FK
name	String
logo_url	String?
other_images_urls	String[]?
level	SponsorLevel
validated	Boolean
validation_token	String?
created_at	DateTime

MerchItem

Field	Type
item_id	UUID PK
event_id	UUID FK
name	String
description	Text
photo_url	String?
other_photos_urls	String[]?
price	Decimal
stock	Integer
active	Boolean

MerchPurchase

Field	Type
order_id	UUID PK
item_id	UUID FK
event_id	UUID FK
visitor_id	UUID FK
quantity	Integer
transaction_id	UUID FK
created_at	DateTime

EventGallery

Field	Type
item_id	UUID PK
event_id	UUID FK
media_url	String
media_type	MediaType
tag	GalleryTag
edition	String?
edition_date	DateTime?

EventResource

Field	Type
resource_id	UUID PK
event_id	UUID FK
title	String
file_url	String

access	ResourceAccess
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EventReport

Field	Type
report_id	UUID PK
event_id	UUID FK
complaint_type	EventComplaintType
description	Text
attachments_urls	String[]
ticket_id	UUID? FK
created_at	DateTime

Enums

```

EventStatus      = DRAFT | PENDING | PUBLISHED | UPCOMING | ONGOING | ENDED | S
EventMode        = TICKETING | SHOWCASE | REGISTRATION_FORM
TicketCategory   = FREE | STANDARD | VIP | EARLY_BIRD | PROMO_CODE | GROUP_PACK
TicketStatus     = VALID | USED | CANCELLED | REFUNDED
DeliveryChannel   = EMAIL | WHATSAPP
SponsorLevel     = PLATINUM | GOLD | SILVER | BRONZE | MEDIA_PARTNER
ResourceAccess    = FREE | AFTER_TICKET | AFTER_REGISTRATION
ScanMethod       = CAMERA | MANUAL_ENTRY
ScanResult       = VALID | ALREADY_USED | INVALID
EventComplaintType = TICKET_NOT_RECEIVED | INVALID_QR | REFUND_ISSUE | ENTRY_PROB

```

Relations

- Event 1 → 1 EventVenue
- Event 1 → * TicketType
- Event 1 → * PromoCode
- TicketType 1 → * TicketPurchase
- Event 1 → * TemporaryReservation
- TicketPurchase 1 → * EntryScan

- Event 1 → * ProgramSession
- Event 1 → * Speaker
- ProgramSession * → 0..1 Speaker
- Event 1 → * EventSponsor
- Event 1 → * MerchItem
- MerchItem 1 → * MerchPurchase
- Event 1 → * EventGallery
- Event 1 → * EventResource
- Event 1 → * EventReport
- Event 1 → 1 RefundPolicy

DC-05 — Donations & Crowdfunding v2

Fundraiser

Field	Type
fundraiser_id	UUID PK
user_id	UUID FK
slug	String unique
title	String
description	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
goal_amount	Decimal?
collected_amount	Decimal

donors_count	Integer
close_at	DateTime?
status	FundraiserStatus
suggested_amounts	Decimal[]
minimum_amount	Decimal
show_donors	Boolean
show_amounts	Boolean
show_latest_donation	Boolean
comments_enabled	Boolean
created_at	DateTime

Donation

Field	Type
donation_id	UUID PK
fundraiser_id	UUID FK
visitor_id	UUID FK
user_id	UUID? FK
transaction_id	UUID FK
amount	Decimal
donor_name	String
email	String?
anonymous	Boolean
support_message	Text?
receipt_channel	DeliveryChannel?
status	DonationStatus

short_id	String(8)
created_at	DateTime

FundraiserComment

Field	Type
comment_id	UUID PK
fundraiser_id	UUID FK
name	String
message	Text
status	CommentStatus
created_at	DateTime

FundraiserUpdate

Field	Type
update_id	UUID PK
fundraiser_id	UUID FK
title	String
body	Text
photos_urls	String[]
videos_urls	String[]?
published_at	DateTime

CFProject

Field	Type
project_id	UUID PK

user_id	UUID FK
slug	String unique
title	String
description	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
goal_amount	Decimal
raised_amount	Decimal
backers_count	Integer
deadline	DateTime
status	CFStatus
created_at	DateTime

Tier

Field	Type
tier_id	UUID PK
project_id	UUID FK
name	String
min_amount	Decimal
description	Text
quota_max	Integer?
backers_count	Integer
active	Boolean

TierOption

Field	Type
option_id	UUID PK
tier_id	UUID FK
label	String
available_values	String[]

CFContribution

Field	Type
contribution_id	UUID PK
project_id	UUID FK
tier_id	UUID? FK
visitor_id	UUID FK
user_id	UUID? FK
transaction_id	UUID FK
amount	Decimal
contributor_name	String
email	String
country	String
chosen_options	JSON?
message	Text?
status	ContribStatus
short_id	String(8)
created_at	DateTime

CFTeamMember

Field	Type
member_id	UUID PK
project_id	UUID FK
name	String
role	String
bio	Text?
photo_url	String?
other_photos_urls	String[]?
social_links	JSON?

CFUpdate

Field	Type
update_id	UUID PK
project_id	UUID FK
title	String
body	Text
photos_urls	String[]
videos_urls	String[]?
notify_backers	Boolean
published_at	DateTime

CFQuestion

Field	Type
question_id	UUID PK

project_id	UUID FK
question	Text
answer	Text?
answered_at	DateTime?
created_at	DateTime

CFComment

Field	Type
comment_id	UUID PK
project_id	UUID FK
name	String
message	Text
status	CommentStatus
created_at	DateTime

Enums

```

FundraiserStatus = DRAFT | PENDING | ACTIVE | CLOSED | CANCELLED
DonationStatus   = PENDING | CONFIRMED | CANCELLED | REFUNDED
CFStatus         = DRAFT | PENDING | ACTIVE | GOAL_REACHED | CLOSED_SUCCESS | CLO
ContribStatus    = PENDING | CONFIRMED | CANCELLED | REFUNDED
CommentStatus    = VISIBLE | HIDDEN | DELETED

```

Relations

- Fundraiser 1 → * Donation
- Fundraiser 1 → * FundraiserComment
- Fundraiser 1 → * FundraiserUpdate
- CFProject 1 → * Tier
- Tier 1 → * TierOption

- CFProject 1 → * CFContribution
 - CFContribution * → 0..1 Tier
 - CFProject 1 → * CFTeamMember
 - CFProject 1 → * CFUpdate
 - CFProject 1 → * CFQuestion
 - CFProject 1 → * CFComment
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DC-06 — Sponsorship, Lotteries & Contests v3

SponsorCall

Field	Type
call_id	UUID PK
user_id	UUID FK
slug	String
title	String
event_description	Text
expected_audience	String
photos_urls	String[]
videos_urls	String[]?
deadline	DateTime
status	CallStatus

SponsorshipLevel

Field	Type
level_id	UUID PK

call_id	UUID FK
name	LevelName
min_amount	Decimal
perks	Text
available	Boolean

SponsorApplication

Field	Type
application_id	UUID PK
call_id	UUID FK
company_name	String
email	String
phone	String
sector	String
level_id	UUID FK
description	Text
logo_url	String?
other_images_urls	String[]?
documents_urls	String[]
transaction_id	UUID FK
status	AppStatus
decided_at	DateTime?
created_at	DateTime

Contest

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Field	Type
contest_id	UUID PK
user_id	UUID FK
slug	String
title	String
description	Text
rules	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
format	ContestFormat
entry_fee	Decimal?
max_participants	Integer?
registration_opens_at	DateTime
registration_closes_at	DateTime
finals_at	DateTime
status	ContestStatus
registration_mode	RegistrationMode
cash_prize	Decimal?
escrow_id	UUID? FK
custom_fields	JSON?
created_at	DateTime

Prize

Field	Type
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prize_id	UUID PK
contest_id	UUID? FK
lottery_id	UUID? FK
title	String
description	Text
cash_value	Decimal?
photos_urls	String[]?

ContestParticipant

Field	Type
participant_id	UUID PK
contest_id	UUID FK
visitor_id	UUID FK
user_id	UUID? FK
name	String
email	String
photos_urls	String[]?
videos_urls	String[]?
team	String?
category	String?
score	Decimal
rank	Integer
status	ParticipantStatus
entry_fee_transaction_id	UUID? FK
custom_fields_data	JSON?

created_at	DateTime

Submission

Field	Type
submission_id	UUID PK
participant_id	UUID FK
contest_id	UUID FK
files_urls	String[]
photos_urls	String[]?
videos_urls	String[]?
text_content	Text?
quiz_answers	JSON?
auto_score	Decimal?
manual_score	Decimal?
submitted_at	DateTime

BracketMatch

Field	Type
match_id	UUID PK
contest_id	UUID FK
round	Integer
participant_a_id	UUID FK
participant_b_id	UUID FK
winner_id	UUID? FK

score_a	Decimal?
score_b	Decimal?
completed	Boolean

EscrowDeposit

Field	Type
escrow_id	UUID PK
campaign_id	UUID
campaign_type	String
user_id	UUID FK
amount	Decimal
deposit_tx_id	UUID FK
status	EscrowStatus
winner_id	UUID? FK
released_at	DateTime?
created_at	DateTime

PrizePayment

Field	Type
payment_id	UUID PK
escrow_id	UUID FK
winner_id	UUID FK
payout_phone	String
payout_tx_id	UUID FK
status	PayoutStatus

created_at	DateTime
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Lottery

Field	Type
lottery_id	UUID PK
user_id	UUID FK
event_id	UUID? FK
slug	String
title	String
description	Text
cover_photo_url	String?
additional_photos_urls	String[]
videos_urls	String[]?
ticket_price	Decimal
max_tickets	Integer
tickets_sold	Integer
draw_at	DateTime
draw_mode	DrawMode
status	LotteryStatus
draw_completed	Boolean
created_at	DateTime

LotteryTicket

Field	Type
ticket_id	UUID PK

lottery_id	UUID FK
unique_number	String
buyer_name	String
buyer_email	String
buyer_phone	String
visitor_id	UUID FK
transaction_id	UUID FK
status	LotteryTicketStatus
created_at	DateTime

LotteryDraw

Field	Type
draw_id	UUID PK
lottery_id	UUID FK
winning_numbers	String[]
drawn_at	DateTime
triggered_by	UUID

Enums

CallStatus

= ACTIVE | CLOSED

LevelName

= PLATINUM | GOLD | SILVER | BRONZE

AppStatus

= PENDING | APPROVED | REJECTED

ContestFormat

= SUBMISSION | QUIZ | BRACKET | OTHER

ContestStatus

= DRAFT | ACTIVE | ONGOING | ENDED | CANCELLED

RegistrationMode

= AUTO | MANUAL

ParticipantStatus

= REGISTERED | QUALIFIED | ELIMINATED | WINNER

EscrowStatus

= LOCKED | RELEASED | REFUNDED

PayoutStatus

= PENDING | COMPLETED | FAILED

DrawMode

= AUTO | MANUAL

LotteryStatus

= ACTIVE | CLOSED | CANCELLED

```
LotteryTicketStatus = PENDING_DRAW | WINNER | NON_WINNER
```

Relations

- SponsorCall 1 → * SponsorshipLevel
- SponsorCall 1 → * SponsorApplication
- Contest 1 → * Prize
- Contest 1 → * ContestParticipant
- Contest 1 → * Submission
- Contest 1 → * BracketMatch
- Contest 1 → 0..1 EscrowDeposit
- EscrowDeposit 1 → 0..1 PrizePayment
- Lottery 1 → * Prize
- Lottery 1 → * LotteryTicket
- Lottery 1 → 0..1 LotteryDraw
- Lottery * → 0..1 Event

DC-07 — Design & Customization v2

PageConfig

Field	Type
config_id	UUID PK
campaign_id	UUID
campaign_type	String
blocks	JSON
version	Integer
published	Boolean

saved_at	DateTime
published_at	DateTime?

Block

Field	Type
block_id	UUID PK
config_id	UUID FK
type	BlockType
position	Integer
content	JSON
style	JSON
visible_mobile	Boolean
visible_desktop	Boolean

BrandSettings

Field	Type
brand_id	UUID PK
campaign_id	UUID
primary_color	String
secondary_color	String
accent_color	String
logo_url	String?
heading_font	String
body_font	String
favicon_url	String?

CustomDomain

Field	Type
domain_id	UUID PK
campaign_id	UUID
domain_name	String
cname_target	String
dns_status	DNSStatus
ssl_active	Boolean
verified_at	DateTime?
created_at	DateTime

DesignTemplate

Field	Type
template_id	UUID PK
name	String
campaign_type	String
preview_url	String
config_json	JSON
credit_cost	Integer
active	Boolean

TicketDesign

Field	Type
design_id	UUID PK
event_id	UUID FK

elements	JSON
preview_url	String?
active	Boolean

IACredit

Field	Type
credit_id	UUID PK
user_id	UUID FK
balance	Integer
initial_free_credits	Integer
updated_at	DateTime

IAGeneration

Field	Type
generation_id	UUID PK
user_id	UUID FK
campaign_id	UUID
prompt	Text
generated_config	JSON
credits_used	Integer
created_at	DateTime

IACreditPack

Field	Type
pack_id	UUID PK

name	String
nb_credits	Integer
price	Decimal
active	Boolean

Enums

```
BlockType = BANNER | RICH_TEXT | IMAGE | GALLERY | VIDEO | COUNTDOWN | CANDIDATES
DNSStatus = PENDING | VERIFIED | ERROR
```

Relations

- PageConfig 1 → * Block
- PageConfig * → 0..1 BrandSettings
- PageConfig * → 0..1 CustomDomain
- IACredit 1 → * IAGeneration

DC-08 — Finance & Payments MVP

Transaction

Field	Type
transaction_id	UUID PK
campaign_id	UUID
campaign_type	String
type	TxType
visitor_id	UUID FK
user_id	UUID? FK
gross_amount	Decimal
moledi_commission	Decimal

psp_fee	Decimal
net_organizer	Decimal
status	TxStatus
idempotency_key	String unique
aggregator_id	UUID FK
external_tx_id	String
country	String
payment_method	PaymentMethod
operator	String?
webhook_payload	JSON?
correlation_id	String
applied_commission_config_id	UUID? FK
initiated_at	DateTime
confirmed_at	DateTime?
expires_at	DateTime?

UserCommissionConfig

Field	Type
config_id	UUID PK
user_id	UUID? FK
campaign_type	String?
vote_rate	Decimal?
standard_rate	Decimal?
active	Boolean

note	String?
configured_by	UUID FK
valid_from	DateTime
valid_until	DateTime?
updated_at	DateTime

Commission lookup order:

1. UserCommissionConfig where `user_id + campaign_type` match
2. UserCommissionConfig where `user_id` match, `campaign_type` IS NULL
3. Global CommissionConfig

CommissionConfig

Field	Type
config_id	UUID PK
type	CommissionType
rate	Decimal
floor_amount	Decimal?
active_from	DateTime
modified_by	UUID FK

PayoutRequest

Field	Type
payout_id	UUID PK
user_id	UUID FK
requested_amount	Decimal
payout_fee	Decimal

net_amount	Decimal
payout_phone	String
aggregator_id	UUID FK
status	PayoutStatus
tracking_ref	String?
proof_url	String?
requested_at	DateTime
executed_at	DateTime?
failure_reason	String?

OrganizerBalance

Field	Type
balance_id	UUID PK
user_id	UUID FK
available_amount	Decimal
reserved_amount	Decimal
updated_at	DateTime

Aggregator

Field	Type
aggregator_id	UUID PK
name	String
countries	String[]
payment_methods	PaymentMethod[]
priority	Integer

active	Boolean
api_endpoint	String
health_endpoint	String
timeout_seconds	Integer
operational_status	AggStatus
success_rate	Decimal
avg_latency_ms	Integer

AggregatorHealthLog

Field	Type
log_id	UUID PK
aggregator_id	UUID FK
success	Boolean
latency_ms	Integer
checked_at	DateTime

FailoverLog

Field	Type
failover_id	UUID PK
transaction_id	UUID FK
source_agg_id	UUID FK
target_agg_id	UUID FK
reason	String
logged_at	DateTime

WebhookLog

Field	Type
webhook_log_id	UUID PK
aggregator_id	UUID FK
transaction_id	UUID? FK
raw_payload	JSON
signature_valid	Boolean
processing_status	String
received_at	DateTime

PSPReconciliation

Field	Type
recon_id	UUID PK
aggregator_id	UUID FK
reconciled_date	Date
transactions_checked	Integer
discrepancies_count	Integer
discrepancy_amount	Decimal
report_json	JSON
generated_at	DateTime
alert_sent	Boolean

CountryConfig

Field	Type
country_code	String PK
country_name	String

active	Boolean
currency	String
aggregator_ids	UUID[]
methods_available	PaymentMethod[]
updated_at	DateTime

MinimumPayoutConfig

Field	Type
config_id	UUID PK
user_id	UUID? FK
min_amount	Decimal
updated_at	DateTime

PayoutBlock

Field	Type
block_id	UUID PK
user_id	UUID? FK
active	Boolean
reason	String
blocked_by	UUID FK
blocked_at	DateTime
unblocked_at	DateTime?
unblocked_by	UUID? FK

Enums

TxType = VOTE | TICKET | DONATION | CF_CONTRIBUTION | LOTTERY_TICKET | CO
TxStatus = INITIATED | PENDING | CONFIRMED | FAILED | EXPIRED | REFUNDED |
PaymentMethod = MOBILE_MONEY | CARD | PAYPAL
AggStatus = OPERATIONAL | DEGRADED | DOWN
PayoutStatus = PENDING | COMPLETED | FAILED
CommissionType = VOTE | TICKET | DONATION | CF | CONTEST | LOTTERY | ESCROW_FEE |

Relations

- Transaction * → 1 Aggregator
- Transaction * → 0..1 UserCommissionConfig
- Transaction * → 0..1 CommissionConfig
- Transaction 1 → * FailoverLog
- Transaction 1 → * WebhookLog
- Aggregator 1 → * AggregatorHealthLog
- Aggregator 1 → * PSPReconciliation
- PayoutRequest * → 1 Aggregator
- OrganizerBalance 1 → 1 User

DC-09 — Support, Notifications & Audit MVP

SupportTicket

Field	Type
ticket_id	UUID PK
number	String unique
visitor_id	UUID FK
user_id	UUID? FK
campaign_id	UUID?
campaign_type	String?

subject	TicketSubject
description	Text
status	TicketStatus
priority	TicketPriority
assigned_to	UUID? FK
created_at	DateTime
updated_at	DateTime

TicketMessage

Field	Type
message_id	UUID PK
ticket_id	UUID FK
author_id	UUID FK
author_role	String
content	Text
attachments_urls	String[]
created_at	DateTime

Notification

Field	Type
notif_id	UUID PK
user_id	UUID FK
type	NotifType
title	String
message	Text

read	Boolean
entity_id	UUID?
entity_type	String?
created_at	DateTime

NotificationLog

Field	Type
log_id	UUID PK
notif_id	UUID FK
channel	NotifChannel
send_status	SendStatus
sent_at	DateTime

MassEmailCampaign

Field	Type
campaign_id	UUID PK
admin_id	UUID FK
subject	String
body_html	Text
segment	JSON
recipients_count	Integer
delivery_rate	Decimal?
sent_at	DateTime

MassWhatsAppCampaign

--	--

Field	Type
campaign_id	UUID PK
admin_id	UUID FK
message	Text
segment	JSON
recipients_count	Integer
green_api_status	GreenAPIStatus
sent_at	DateTime

AuditLog

Field	Type
audit_id	UUID PK
actor_id	UUID
actor_role	String
action_type	AuditAction
entity_type	String
entity_id	UUID
before_value	JSON?
after_value	JSON?
ip	String
timestamp	DateTime

AuditLog is immutable — NO UPDATE or DELETE ever permitted on this table.

PlatformAlert

Field	Type

alert_id	UUID PK
type	AlertType
severity	Severity
message	Text
entity_id	UUID?
resolved	Boolean
created_at	DateTime

AnomalyDetection

Field	Type
anomaly_id	UUID PK
poll_id	UUID? FK
type	AnomalyType
description	Text
occurrences	Integer
status	AnomalyStatus
detected_at	DateTime
resolved_at	DateTime?

Enums

```

TicketSubject  = VOTE | TICKET | PAYMENT | REFUND | FRAUD | TECHNICAL | OTHER
TicketStatus   = OPEN | IN_PROGRESS | RESOLVED | CLOSED
TicketPriority = LOW | NORMAL | HIGH | CRITICAL
NotifType      = ADMIN_MESSAGE | CAMPAIGN_VALIDATED | CAMPAIGN_REJECTED | CAMPAIG
NotifChannel   = EMAIL | WHATSAPP | IN_APP
SendStatus     = SENT | FAILED | PENDING
GreenAPIStatus = CONNECTED | DISCONNECTED
AuditAction    = CREATE | UPDATE | DELETE | VALIDATE | REJECT | SUSPEND | CLOSE |
AlertType      = LONG_PENDING_TX | AGGREGATOR_DOWN | GREEN_API_DOWN | SUSPICIOUS_
Severity       = INFO | WARNING | CRITICAL

```

```
AnomalyType      = VOTE_SPIKE | IDENTICAL_PAYMENTS | MULTIPLE_DEVICES | ABNORMAL_RH
AnomalyStatus    = DETECTED | UNDER_INVESTIGATION | RESOLVED | FALSE_POSITIVE
```

Relations

- SupportTicket 1 → * TicketMessage
- Notification 1 → * NotificationLog

DC-10 — Back-office & Global Config MVP

SiteConfig

Field	Type
config_id	UUID PK
contact_email	String
admin_email	String
support_phone	String
whatsapp_support	String
social_links	JSON
response_time_label	String
updated_at	DateTime

GlobalFAQ

Field	Type
faq_id	UUID PK
type	FAQType
question	String
answer	Text

position	Integer
active	Boolean

FeatureFlag

Field	Type
flag_id	UUID PK
name	String
description	String
active	Boolean
modified_by	UUID FK
updated_at	DateTime

MaintenanceMode

Field	Type
mode_id	UUID PK
active	Boolean
message	Text
eta_return	DateTime?
activated_by	UUID FK
activated_at	DateTime?

KPISnapshot

Field	Type
kpi_id	UUID PK
type	KPIType

value	Decimal
period	String
computed_at	DateTime

CannedReply

Field	Type
reply_id	UUID PK
subject	TicketSubject
title	String
content	Text
active	Boolean

SystemEmailTemplate

Field	Type
template_id	UUID PK
type	String
subject	String
body_html	Text
variables	String[]
updated_at	DateTime

LegalPage

Field	Type
page_id	UUID PK
type	LegalType

content	Text
updated_at	DateTime
updated_by	UUID FK

LanguageConfig

Field	Type
language_id	UUID PK
code	Language
active	Boolean
is_default	Boolean

Enums

```
FAQType    = HOW_IT_WORKS | POLL_TEMPLATE | EVENT_TEMPLATE | DONATION_TEMPLATE | C
LegalType  = TERMS | SALES_TERMS | LEGAL_NOTICE | COOKIES | PRIVACY
KPIType    = MAU | ORGANIZER_RETENTION | CONVERSION_RATE | MEDIAN_VOTE_TIME | MONT
```

Component Diagram — Global Architecture

Layer 1 — Clients

Component	Details
Web Frontend	React/Vue SPA — Visitor & Organizer — mobile-first < 500KB vote page
Admin Back-office	React — /admin/* routes — JWT + 2FA — session expires 2h

Layer 2 — API & Business Services (Node.js)

Component	Methods
REST API	Public + Organizer + Admin routes — Webhook endpoints POST /webhook/payment/:provider

PaymentService	charge() · refund() · payout()
VoteService	confirmPayment() · validateAntiDup() · registerVote()
TicketService	activateTicket() · validateQR() · generatePDF()
NotificationService	sendEmail() · sendWhatsApp() · createInApp()
AuditService	log() — append-only
StorageService	upload() · generateSignedUrl()
AIService	generatePage() · deductCredits()

Layer 3 — Payment Gateways

Component	Details
Gateway A/B/N	/lib/payment/gateways/ — interface: initiatePayment() · checkStatus() · refund() — timeout 10s
Aggregator Selector	selectAggregator(country, method) — reads Cache health — returns first OPERATIONAL by priority
Failover Manager	on primary fail → DEGRADED 5min → retry on next — logs to FailoverLog
aggregators.yaml	/config/payment/ — hot-reload via /admin/payment/reload-config — no redeploy needed

Layer 4 — Async Jobs (node-cron)

Job	Frequency	Action
expire-pending	every 10 min	Expires tx > 30min — auto-refund if funds taken
reconcile	daily overnight	Compares DB vs PSP — alerts admin on gaps
health-check	every 2 min	Pings each aggregator — updates Cache — alerts on 3 consecutive failures

Layer 5 — Data

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Component	Details
Primary Database	PostgreSQL — all DC-01→DC-10 — AuditLog append-only
Read Replica	Admin analytics only — no load on primary
Cache (Redis)	payment:health:[agg] · payment:aggregators · vote counters — TTL 30s / 5min
File Storage (S3)	Photos · PDFs · PV · Attachments — signed URLs

Layer 6 — External Services

Service	Usage
Green API	WhatsApp notifications · OTP · mass messaging
SMTP	Transactional emails · mass email campaigns
Google Maps API	Embedded maps on event pages
Google OAuth2	Social login — email auto-verified (V1+)
Ad Network	Video ads in vote tunnel — 30% Moledi / 70% organizer (V2+)
Anthropic Claude API	AI page generation — JSON block config (V4+)

Key Communication Flows

Flow 1 — Paid vote Client → API → VoteService → PaymentService → selectAggregator() → Gateway → PSP PSP → POST /webhook/payment/:provider → HMAC check → confirmPayment() → DB update → Notification

Flow 2 — Transaction expiry Job expire-pending → checkStatus() → Gateway → if expired: refund() → DB → Notification

Flow 3 — Automatic failover charge() → Primary fails → DEGRADED in Cache 5min → selectAggregator() picks next → retry once

Flow 4 — Organizer payout POST /api/payments/payout → checks (balance · block · minimum · phone) → debit reserved → Gateway → webhook → COMPLETED or FAILED + restore balance

Flow 5 — Closing report Admin closes poll → AuditLog → async job → assemble results → SHA-256 → upload PDF → ClosingReport in DB

Flow 6 — Health monitoring Job every 2min → GET health_endpoint each aggregator → update Cache → 3 failures: PlatformAlert + admin notification

Summary

Type	Count	Files
Class diagrams	10	DC01 → DC10 .puml
Component diagram	1	COMP_Architecture.puml
Total	11	

DC	Sub-diagram	Classes	Version
01	Identity & Session	5	MVP
02	Authentication & Users	9	MVP
03	Polls & Votes	17	MVP
04	Events & Ticketing	15	V1
05	Donations & Crowdfunding	12	V2
06	Sponsorship, Lotteries & Contests	12	V3
07	Design & Customization	9	V2
08	Finance & Payments	13	MVP
09	Support, Notifications & Audit	9	MVP
10	Back-office & Global Config	9	MVP
Total		110 classes	